Aditya Kaushik

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B.E in C.S.E
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Education

- **RV Institute Of Technology and Management, Bangalore** — *2023–Present*
  B.E in C.S.E — CGPA: 8.95
- **D.P.S Bhilai** — *2021*
  CBSE, Chhattisgarh — Percentage: 93%
- **D.P.S Bhilai** — *2019*
  CBSE, Chhattisgarh — Percentage: 93%

Experience and Personal Projects

- **Text-based Screening for Parkinson's Disease**
  Developed an automated screening system to assess Parkinson's risk from patient symptom descriptions.
  - **Backend (Python + ML Pipeline):** Built a Python pipeline leveraging transformer models on datasets such as PPMI and UCI with functionalities for text preprocessing, feature extraction, and risk scoring. Implemented state-of-the-art pretrained language models (LLMs) for medical screening.
  - **Machine Learning Models:** Utilized Hugging Face Transformers for NLP-based risk detection. Developed robust classification algorithms with clinical accuracy while maintaining low resource overhead.
  - **Deployment & Interface:** Created a Streamlit-based interface for real-time symptom analysis and risk assessment. Enabled automated medical screening with deployable text screening modules.
  - **Data & Performance:** Processed and analyzed medical datasets from multiple sources (PPMI, UCI, Kaggle) with preprocessing pipelines for optimal model performance and reliable classification results.
  - Tools & Technologies Used: Python, Hugging Face Transformers, scikit-learn, Streamlit, PPMI/UCI/Kaggle datasets.

- **Interactive Stock Tracking Web Application**
  Built a full-stack virtual stock trading platform with real-time price updates, portfolio management, and competitive leaderboard.
  - **Backend (Node.js + Express.js):** Built a RESTful API with 25+ endpoints for authentication, trading, portfolio management, and admin operations. Implemented JWT-based authentication with role-based access control. Designed PostgreSQL database schema with Prisma ORM, managing 6 interconnected tables. Deployed on Vercel with serverless functions.
  - **Frontend (React + Vite):** Developed a responsive Single-Page Application (SPA) with React Router. Created 15+ reusable components including real-time stock tables and interactive trade modals. Implemented Context API for global state management. Styled with Tailwind CSS for a modern, mobile-responsive UI/UX.
  - **Database & Infrastructure:** Utilized PostgreSQL database hosted on Prisma Cloud with 100+ stock records. Configured automated database migrations and seeding scripts. Implemented connection pooling and query optimization for sub-100ms transaction response times.
  - Tools & Technologies Used: React, Node.js, Express.js, PostgreSQL, Prisma ORM, JWT, Vercel, Tailwind CSS.

- **Real-Time Payment Fraud Detection Engine**
  Developed a high-throughput, low-latency financial risk engine to detect fraudulent transactions in real-time streams using Graph Neural Networks (GNN) and event-driven architecture.
  - **AI & Machine Learning Architecture:** Engineered a Graph Neural Network (GNN) using PyTorch Geometric to model complex transaction relationships, identifying organized fraud rings and synthetic identities. Tackled extreme class imbalance (0.1% fraud rate) using SMOTE sampling and Focal Loss functions, improving model AUPRC over baseline logistic regression. Implemented model explainability using SHAP values to generate transparent "reason codes" for every flagged transaction.
  - **System Engineering & Performance (Low-Latency):** Built a real-time ingestion pipeline using Apache Kafka for event streaming and Redis for feature caching, processing 10,000+ transactions per second (TPS). Optimized model inference speed by converting models to ONNX Runtime and deploying on AWS Lambda, achieving end-to-end latency of <45ms for live authorization. Containerized microservices using Docker and orchestrated with Kubernetes for auto-scaling during peak transaction volumes.
  - **Results & Impact:** Reduced False Positive Rate (FPR) while maintaining real-time authorization speeds. Successfully identified complex fraud patterns through graph-based analysis of transaction networks. Achieved regulatory compliance with transparent, explainable AI decisions for audit trails.
  - Tools & Technologies Used: Python, PyTorch, Apache Kafka, Redis, AWS (Lambda/S3), Docker, Kubernetes, ONNX, SQL, Graph Neural Networks.

- **Aniverse — AI-Powered Anime Discovery Platform**
  A next-generation anime database and discovery engine featuring semantic "scene-to-anime" search and an agentic AI assistant for automated list management.

- **Intelligent Search & ML Architecture:** Developed a multimodal search engine allowing users to find anime via natural language scene descriptions (e.g., "fight in the rain with blue fire"). Leveraged **OpenAI CLIP** or **Sentence-Transformers** to map text and image embeddings into a shared vector space, stored in a **Pinecone** vector database for millisecond-latency semantic retrieval.
- **Agentic AI Assistant:** Engineered a specialized AI agent using **LangGraph** and **Function Calling** that acts as a personal "Assistant." Enabled the agent to autonomously execute CRUD operations—adding, rating, or removing titles from a user's watchlist—based on a single natural language prompt
- **Frontend UX/UI:** Architected a modern, high-performance SPA using **Next.js 15** and **Framer Motion** for fluid animations. Designed a modern interface with skeleton loaders, infinite scrolling, and dynamic glassmorphism components, significantly outperforming legacy platforms (MAL/Anilist) in user engagement metrics.
- Tools & Technologies Used: Next.js, TypeScript, PostgreSQL, Pinecone (Vector DB), LangChain/LangGraph, OpenAI API, Tailwind CSS, Framer Motion.

Technical Skills and Interests

Languages: Python, C, C++, Java, JavaScript, HTML, CSS
Developer Tools: Git, VS Code, Blender, Vercel, Jupyter, Notebook, Linux CLI
Frameworks: React, Node.js, Three.js, NumPy, Pandas
Cloud/Databases: MongoDB, MySQL
Soft Skills: Teaching, Research Writing, Teamwork, Public Speaking, Problem Solving, Workshop Facilitation
Coursework: Data Structures & Algorithms, Machine Learning, DBMS, OS, Computer Networks, Probability and Statistics
Areas of Interest: Machine Learning, Web Development, Data Structures and Algorithms, UI/UX Design

Positions of Responsibility

- **Member** Speakers Club *3 years*
- **Senior Board Member** DSA Club *1 year*
- **Senior Board Member** Entrepreneurship Cell *1 year*
- **Senior Board Member** Webdev Club *1 year*

Achievements

- **Published a Paper** Published a research paper in HBRP Publication on the topic: *Optimizing Soyabean Production: A Data Driven Machine Learning Framework.* *03/2025*
- **Published a Paper** Published a research paper in HBRP Publication on the topic: *Unveiling India's Demographic Mosaic: An Exploratory Data Analysis of Indian Census Data.* *11/2025*